DATA SCIENCE | PAKISTAN POPULATION DATA

What Pakistan's Population Data Taught Me as a Data Fresher

528 sub-divisions, one profiling report, and lessons that go beyond the numbers

528	204.4M	+75.8%
sub-divisions analyzed	people in the latest counts	urban growth since 1998

Pakistan's population is one of the most talked-about numbers in the country. But how often do we look at what the data behind it actually contains?

I ran a data profiling report on a dataset of **528 sub-divisions (tehsils)** across Pakistan, with rural and urban population figures, sex ratios, household sizes, and growth rates. I'm a fresher in data science, so here is what I found, in simple words.

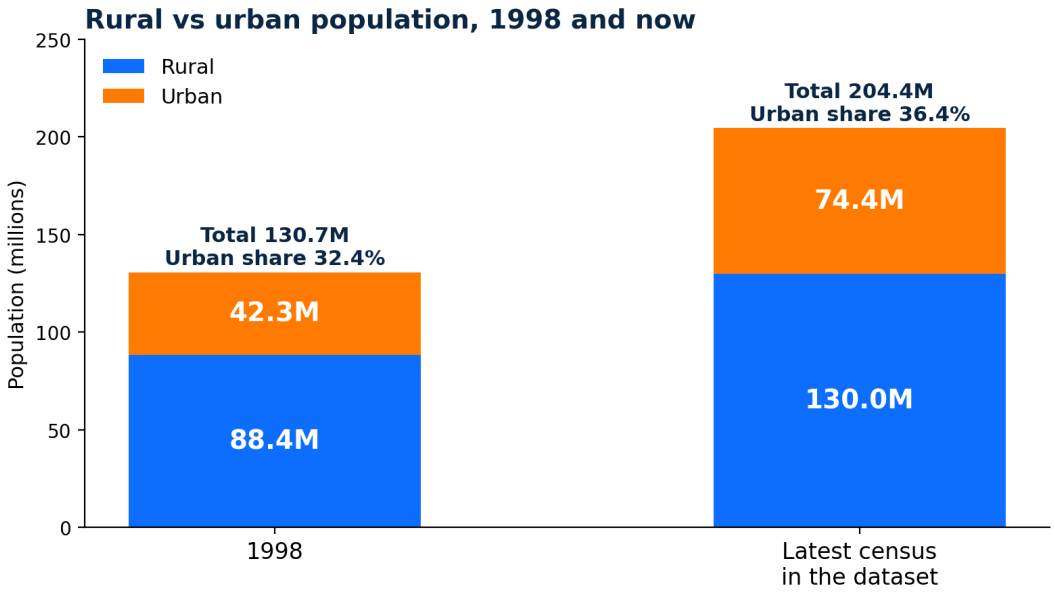
The Dataset at a Glance

- **528 rows**, one for each sub-division
- **21 columns:** province, division, district, sub-division, area, and rural and urban figures for population, male, female, transgender, sex ratio, household size, 1998 population, and annual growth rate
- **5 province labels, 28 divisions, 131 districts**
- **0 missing cells and 0 duplicate rows**

Zero missing values sounds perfect. As you will see, it hides some important details.

Finding 1: Pakistan Is Still Mostly Rural, But Cities Are Growing

Adding up the columns gives **130.0 million rural** and **74.4 million urban** people, about **204.4 million** in total. In 1998 the same columns add up to **88.4 million rural** and **42.3 million urban**, about 130.7 million.



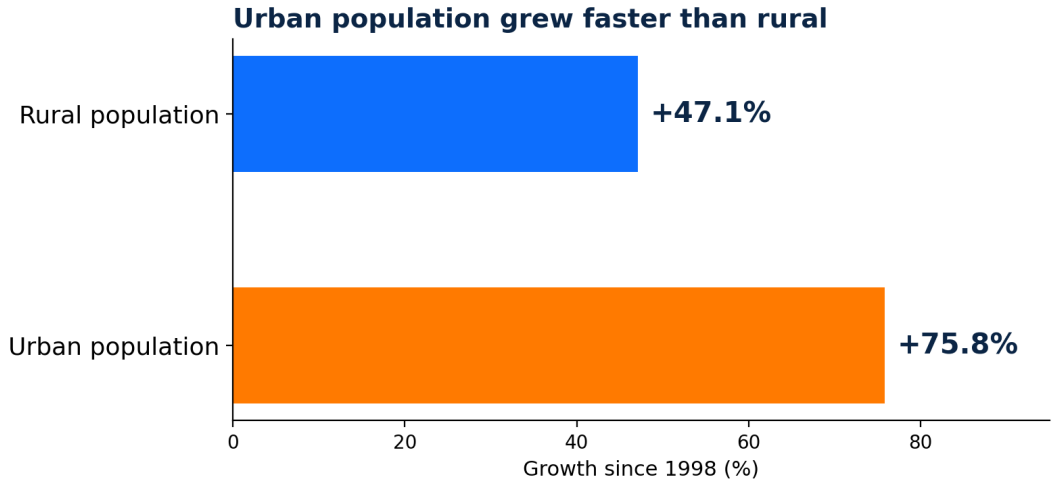
Rural vs urban population, 1998 and the latest counts

The urban share rose from **32.4% to 36.4%**. More than a third of the population now lives in urban areas.

Finding 2: Urban Areas Grew Much Faster

Since 1998, the total population grew by about **56.4%**. But the two groups did not grow equally:

- **Urban population: +75.8%**
- **Rural population: +47.1%**



Growth since 1998, rural vs urban

> **Insight:** **urban areas grew about 1.6 times as fast as rural areas.**

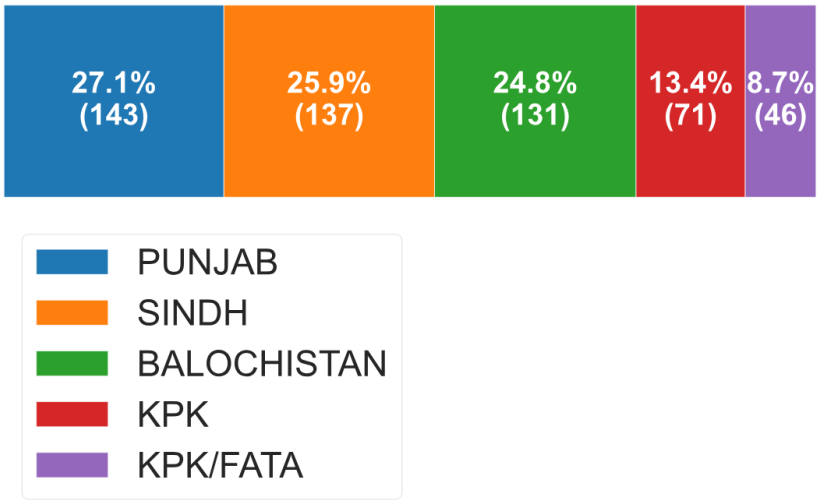
Finding 3: Men Slightly Outnumber Women

In rural areas there are about **66.1 million men and 63.9 million women**, which is roughly **103.5 men per 100 women**. In urban areas the gap is wider: **38.5 million men and 35.9 million women**, about **107.1 men per 100 women**.

The dataset also counts **21,358 transgender people** (9,596 rural and 11,762 urban). Having this category in the data is a good reminder that the groups we choose to count decide who is visible in the analysis.

Finding 4: Provinces Look Balanced, But Count Is Not People

The number of sub-divisions per province looks like this: **Punjab 143** (27.1%), **Sindh 137** (25.9%), **Balochistan 131** (24.8%), **KPK 71** (13.4%), and **KPK/FATA 46** (8.7%).

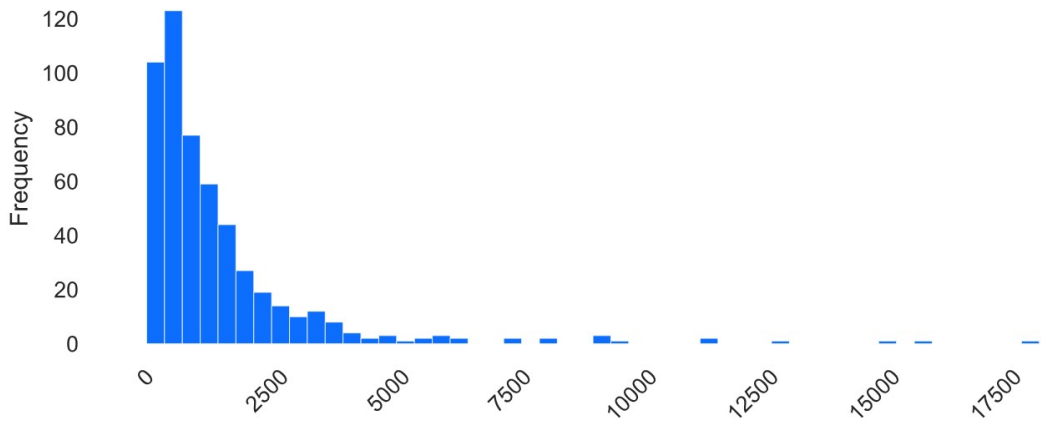


Number of sub-divisions per province

Be careful here. This counts **administrative units, not people**. Balochistan has almost as many sub-divisions as Punjab, but the units are very different in size. Counting rows is not the same as counting population.

Finding 5: A Few Giants Skew the Averages

The average sub-division covers about **1,492 sq km**, but the largest covers **18,374 sq km**. Most are small, and a few are huge.



Area histogram: most sub-divisions are small, a few are very large

Population works the same way. The largest urban sub-division has **3.65 million people**, about **26 times** the average. Whenever you see a long tail like this, the average alone is misleading.

The Biggest Lesson: Zero Is Not the Same as Missing

The report found **0 missing cells**, but still raised **33 alerts**: 16 high correlations, 16 zero-value alerts, and 1 unique-value alert. The zeros were the most interesting:

- **155 sub-divisions (29.4%)** have **zero urban population**. They are fully rural.
- **34 sub-divisions (6.4%)** have **zero rural population**. They are fully urban.

These zeros are not errors. They mean "this does not exist here." But if you average the urban population across all 528 rows, those zeros pull the number down and give you a wrong picture. The same goes for sex ratio and household size, which show as 0.00 where there is no population.

> **Lesson: decide whether a zero means "zero" or "not applicable" before you calculate anything.**

More Things I Checked

- **The totals add up.** In both rural and urban data, male + female + transgender equals the all-sexes total exactly. That is a good sign the data is consistent.
- **Correlated by design.** Sixteen columns are flagged as highly correlated. This is expected, because male, female and total populations move together. It does not mean a hidden pattern.
- **Suspicious values.** The maximum rural annual growth rate is **100%**, and the maximum urban sex ratio is **297.8**. Values like these need checking before you trust them.
- **Old labels.** "KPK" and "KPK/FATA" appear as separate provinces. A data analyst has to decide whether to merge them.

One Important Note

The file does not label the year of the main population columns. The totals (about 204 million) look like the **2017 census**, but I am assuming this. The “1998” columns are labeled. Always check the source and time period of a dataset before you share results.

What I Learned as a Fresher

1. **Zero missing values does not mean clean data.** Look at zeros, outliers and labels too.
2. **Check that totals add up.** It is a quick way to test data quality.
3. **Counts are not the same as people.** Know what each row represents.
4. **Skewed data needs more than the average.** Check the maximum and the median too.
5. **Be honest about what you do not know.** If the year or source is unclear, say so.

What’s Next

Next I want to group the data by province and division to compare population, growth and urbanization across regions, and to see which sub-divisions are growing fastest.

If you work with data: **what is the first thing you check when you get a new dataset?** And which province or region would you like me to explore first? Tell me in the comments.

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